Adaptive Hybrid Physics-Informed Neural Networks for Multi-Dimensional Singular Perturbation Problems

Internship Technical Report Draft

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Abstract

To address the optimization challenges and boundary layer "stiff locking" inherent in solving singularly perturbed problems, this report introduces an Adaptive Hybrid Physics-Informed Neural Network (PINN) for time-dependent equations in 1D and 2D space. The architecture leverages asymptotic coordinate stretching to scale individual Multi-Layer Perceptron (MLP) predictions independently for localized left and right boundary layers. The training pipeline combines an adaptive, hybrid Shishkin-mesh sampling methodology with an explicit L-BFGS reactivation mechanism to consistently bypass optimization plateaus. For complex two-parameter problems, the framework incorporates automated, regime-based scaling to maintain stability across shifting physical scales. Comprehensive validation across multiple singular regimes demonstrates that the framework successfully eliminates spectral bias, captures sharp spatial gradients, and accelerates convergence speed compared to standard neural architectures. Ultimately, this unified approach provides a highly precise, robust alternative to traditional neural PDE solvers, establishing a scalable multi-dimensional framework for simulating highly stiff physical systems.

Contents

1 Introduction

The strategic resolution of singularly perturbed dynamical systems is critical in fields such as chemical kinetics, fluid dynamics, and neuroscience, where governing equations are often characterized by multiple small parameters associated with leading differential operators. These parameters induce "stiff" behaviors, resulting in solution profiles that exhibit sharp internal or boundary layers. In these narrow intervals, state variables undergo rapid fluctuations, creating high-contrast multiscale dynamics that are notoriously difficult to resolve.

To bridge the gap between deep learning and classical numerical analysis, Physics-Informed Neural Networks (PINNs) have been widely adopted [13]. PINNs approximate the solution to a given system by treating neural networks as continuous function approximators. The underlying physics are enforced by constructing a multi-objective loss function consisting of mean squared errors from the domain interior (the PDE residual), initial conditions, and boundary constraints. The network is then trained via backpropagation, penalizing any deviations from the physical law.

Standard Physics-Informed Neural Networks (PINNs) exhibit a significant limitation known as spectral bias, a phenomenon in gradient-based optimization where the network prioritized learning low-frequency components of the solution [14]. Consequently, standard architectures fail to capture high-frequency features or the sharp gradients characteristic of stiff systems without excessive computational overhead or manual tuning.

This structural vulnerability necessitates a departure from uniform network architectures and isotropic point sampling. In this work, we present an alternative: an Adaptive Hybrid PINN. This framework directly counters spectral bias by utilizing coordinate-stretching Multi-Layer Perceptrons (MLPs) dedicated to asymptotic boundary resolution. By pairing this architecture with a dynamic Shishkin-mesh point allocation strategy, our solution enforces strict physical compliance exactly where the gradients are steepest.

2 Literature Review

The formulation of structural network layers specifically optimized to handle the mathematical imbalances found in singular boundaries was advanced by Wang and Zhang (2023) [1] who introduced the approach of deploying entirely separate neural networks, which explicitly decouples the smooth background dynamics from the stiff boundary behaviors. Rather than expecting a uniform network to naturally approximate these sharp features, Wang et al. (2024) [2] introduced localized asymptotic scaling within the internal layers to analytically stretch coordinates near sharp gradients, transforming steep drops into smooth, trackable profiles.Further Li et al. (2025) [3], focused on developing a unified weighted-loss physics-informed neural network architecture capable of stabilizing composite objective functions. To better handle these regional variations,

To prevent the spatial starvation of training points within these microscopic regions, modern networks often augment their collocation pipelines with automated point generation strategies. A classical cornerstone for this approach is the Shishkin mesh, established in the foundational texts by Miller et al. (1996) [7] and Farrell et al. (2000) [6], which dictates a strict mathematical layout for piecewise-uniform point distribution based on analytical boundary layer widths. Translating dynamic domain adjustments into deep learning solvers, a modern solution for the one-dimensional Cahn–Hilliard equation developed an improved causal physics-informed neural network featuring an error-based adaptive resampling strategy, which continuously shifts and focuses collocation points onto areas experiencing severe residual spikes mid-flight [12].

When evaluating multi-parameter challenges where competing transport forces occur simultaneously, managing the scaling ratios between independent parameters is critical. Early strategies analyzed by Zhang et al. (2023) [4] utilized a fixed geometric mean of epsilon values to estimate layer widths and balance global derivatives. However, to handle diverse physical domains dynamically, Kumar et al. (2024) [5] and Boro et al. (2025) [10] introduced specialized regimes for two parameters, allowing the framework to modify its internal coordinate scaling based on whether the system is governed by convection-dominated or diffusion-dominated physics.

Even with robust parameter handling, gradient-based optimization routines frequently stall in the highly non-convex loss landscapes typical of singular systems [14]. To combat this optimization plateau, Liu et al. (2024) [9] proposed a reactivating optimization algorithm that reactivates the optimization process after it converges, so that the loss value and accumulative error of the network is reduced further. In parallel to purely mesh-free formulations, the literature notes the brief existence of hybrid frameworks such as FEM-PINN and FB-PINN, which seek to enforce continuous local boundary compliance by merging deep networks with classical finite elements and finite basis functions.

While the reviewed methods address specific, isolated components of numerical stiffness such as custom loss weightings, adaptive point distributions, or individual optimizer restarts, there remains a critical gap in fully integrated neural architectures capable of resolving multi-parameter problems across shifting physical regimes in higher dimensions.

3 Methodology

3.1 Asymptotic Scaling & Regime Assessment

Singular perturbation challenges involve multi-scale physical behaviors where highly localized boundary layers form due to small perturbation parameters ϵ_1 and ϵ_2. The baseline physical governing system modeled in this work is the two-dimensional transient Burgers' equation:

$$u_t - \epsilon_1(u_{xx} + u_{yy}) + \epsilon_2 u(u_x + u_y) + u = 0 \tag{1}$$

To ensure the deep neural network maps these stiff variations without experiencing gradient pathology, asymptotic scaling transforms are parameterized into specific structural regimes. Scaling choices define stretching multipliers s_l and s_r for boundary sub-networks based on domain thresholds:

$$(s_l, s_r) = \begin{cases} (1/\epsilon_1, 1/\epsilon_1) & \text{if } \epsilon_2 = 1, \epsilon_1 \neq 1 \quad \text{(Regime 1 Baseline)} \\ (1/\epsilon_2, \epsilon_2/\epsilon_1) & \text{if } \epsilon_1 = \epsilon_2 \text{ or } \epsilon_1 < \epsilon_2^2 \quad \text{(Regime 2)} \\ (1/\sqrt{\epsilon_1}, 1/\sqrt{\epsilon_1}) & \text{if } \epsilon_2^2 < \epsilon_1 \quad \text{(Regime 3)} \\ (1/\sqrt{\epsilon_1\epsilon_2}, 1/\sqrt{\epsilon_1\epsilon_2}) & \text{otherwise} \end{cases} \tag{2}$$

3.2 Dual-Network Component Decomposition Architecture

To capture global bulk dynamics and sharp localized layers cleanly, the final approximation is structurally decomposed into a baseline regular network and independent singular boundary layer estimators:

$$u_{\text{pred}}(x, y, t) = mlp_r(x, y, t) + mlp_{s,XL}(\sigma_L(x)) + mlp_{s,XR}(\sigma_R(x)) + mlp_{s,YL}(\sigma_L(y)) + mlp_{s,YR}(\sigma_R(y)) \tag{3}$$

where $\sigma_L(z) = \text{sigmoid}(s_l \cdot z)$ and $\sigma_R(z) = \text{sigmoid}(s_r \cdot (z - 1.0))$ define the boundary coordinate transformation operators.

3.3 Shishkin Mesh Point Initialization

Initial allocation uses a structural layout inspired by a piecewise-uniform Shishkin mesh. Transition boundaries (τ_l, τ_r) isolate high-gradient cells from the interior bulk domain using structural log metrics:

$$\tau_l = \min\left(\frac{1}{4}, \frac{2}{s_l}\ln(N_{pde})\right), \quad \tau_r = \min\left(\frac{1}{4}, \frac{2}{s_r}\ln(N_{pde})\right) \tag{4}$$

Collocation points are assigned sequentially: 25% of points map uniformly within the left layer region $[x_{\min}, x_{\min} + \tau_l]$, 25% map within the right layer region $[x_{\max} - \tau_r, x_{\max}]$, and the remaining 50% satisfy the coarse central bulk domain $[x_{\min} + \tau_l, x_{\max} - \tau_r]$.

3.4 Composite PINN Loss Function

The parameter network relies on minimizing a multi-objective loss function consisting of a weighted layout across distinct operational interfaces:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{ic}}\mathcal{L}_{\text{ic}} + \lambda_{\text{b}}\mathcal{L}_{\text{b}} + \mathcal{L}_{\text{f}} \tag{5}$$

where structural settings enforce $\lambda_{\text{ic}} = 10.0$ and $\lambda_{\text{b}} = 10.0$. Individual loss components are calculated directly as:

$$\mathcal{L}_{\text{ic}} = \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} \left| u_{\text{pred}}(x_i^{ic}, y_i^{ic}, t_{\text{start}}) - u_{\text{target}}(x_i^{ic}, y_i^{ic}) \right|^2 \tag{6}$$

$$\mathcal{L}_{\text{b}} = \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} \left| u_{\text{pred}}(x_i^b, y_i^b, t_i^b) \right|^2 \tag{7}$$

$$\mathcal{L}_{\text{f}} = \frac{1}{N_{pde}} \sum_{i=1}^{N_{pde}} \left| u_t^{(i)} - \epsilon_1(u_{xx}^{(i)} + u_{yy}^{(i)}) + \epsilon_2 u^{(i)}(u_x^{(i)} + u_y^{(i)}) + u^{(i)} \right|^2 \tag{8}$$

3.5 Error-Based Adaptive Resampling

To track non-linear structural variations mid-flight, an adaptive, residual-driven generation loop is executed periodically. Let M be a candidate pool drawn uniformly from across the global domain space. The detailed execution steps follow:

Algorithm 1 Error-Based Adaptive Collocation Point Resampling

1: **Input:** Candidate density factor d, number of replacement points N_{adapted}, current network state u_{pred}
2: Generate $N_{\text{candidates}} = N_{\text{adapted}} \times d$ random points $\{x_k, y_k, t_k\}_{k=1}^{N_{\text{candidates}}}$ uniformly across domain
3: Evaluate physical residual $f_k = u_t^{(k)} - \epsilon_1(u_{xx}^{(k)} + u_{yy}^{(k)}) + \epsilon_2 u^{(k)}(u_x^{(k)} + u_y^{(k)}) + u^{(k)}$ for all candidate entries
4: Calculate absolute values $R_k = |f_k|$
5: Construct discrete probability map:

$$P_k = \frac{\sqrt{R_k}}{\sum_{j=1}^{N_{\text{candidates}}} \sqrt{R_j} + 10^{-8}}$$

6: Draw N_{adapted} unique coordinates without replacement from multinomial layout: $\mathcal{S} \sim \text{Multinomial}(\mathbf{P}, N_{\text{adapted}})$
7: **Return:** Adapted collocation coordinates $\{x_s, y_s, t_s\}_{s \in \mathcal{S}}$

3.6 L-BFGS Reactivation Loop

Following first-order exploratory updates via the Adam optimizer, the training strategy introduces a deterministic fine-tuning phase by reactivating a second-order L-BFGS optimizer. To prevent premature stagnation in complex loss landscapes, the tracking threshold is dynamically adjusted at each non-convergent epoch using a scaling factor β, forcing continued gradient exploration:

Algorithm 2 L-BFGS Fine-Tuning Reactivation Loop with β-Scaling

Input: Initialized network parameters $\boldsymbol{\theta}_0$, tolerance $\epsilon = 10^{-5}$, max iterations $E_{\max} = 100$, scaling factor β
Set historical memory size to 50, internal steps per iteration to 20, line search selection to `strong_wolfe`
Initialize tracking variables: $\mathcal{L}_{\text{prev}} \leftarrow 0.0$
for epoch $e = 1$ to $E_{\max}$ **do**
  Define optimization closure tracking loss step $\mathcal{L}_{\text{current}} \leftarrow \mathcal{L}_{\text{total}}(\boldsymbol{\theta})$
  Step parameter weights updates using approximate curvature calculations:

$$\boldsymbol{\theta}_e = \boldsymbol{\theta}_{e-1} - \mathbf{H}_{e-1}^{-1} \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{total}}(\boldsymbol{\theta}_{e-1})$$

  if $e \pmod{10} == 0$ **then**
    Print spatiotemporal operational loss metrics ($\mathcal{L}_{\text{total}}$, $\mathcal{L}_{\text{ic}}$, $\mathcal{L}_{\text{b}}$, $\mathcal{L}_{\text{f}}$)
  end if
  if $\mathcal{L}_{\text{prev}} \neq 0.0$ **then**
    Compute relative change parameter: $\delta = \frac{|\mathcal{L}_{\text{prev}} - \mathcal{L}_{\text{current}}|}{\mathcal{L}_{\text{prev}}}$
    if $\delta < \epsilon$ **then**
      Exit optimization loop due to high-precision convergence
    else
      Update historical trackers: $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}_{\text{current}}$
      Apply reactivation multiplier: $\mathcal{L}_{\text{current}} \leftarrow (1 + \beta) \times \mathcal{L}_{\text{current}}$
    end if
  end if
end for

3.7 Methodology Flowchart

Below is the architectural workflow of the proposed framework:

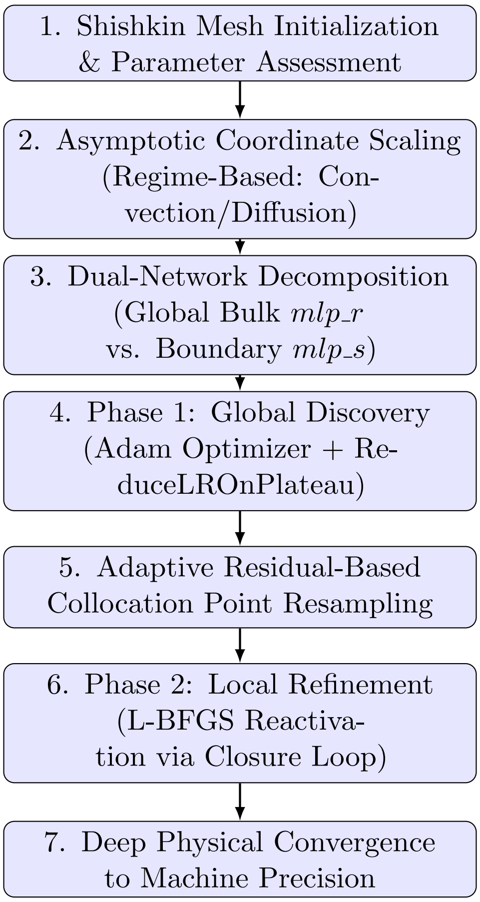

Figure 1: Flowchart illustrating the multi-phase Adaptive Hybrid PINN execution pipeline.

4 Results and Discussions

4.1 Original Numerical Solution Baseline

The exact analytical baseline profile presents a multi-layered spatiotemporal structure under strong diffusion-dominated constraints. The surface mapping illustrates how the solution field forms a prominent, curved peak along the spatial axis that gradually dissipates over time. Because the diffusion parameter dominates the system, the sharp initial discontinuities are smoothed out structurally across the center bulk of the domain, leaving a well-defined, continuous macroscopic wave profile that acts as our validation benchmark.

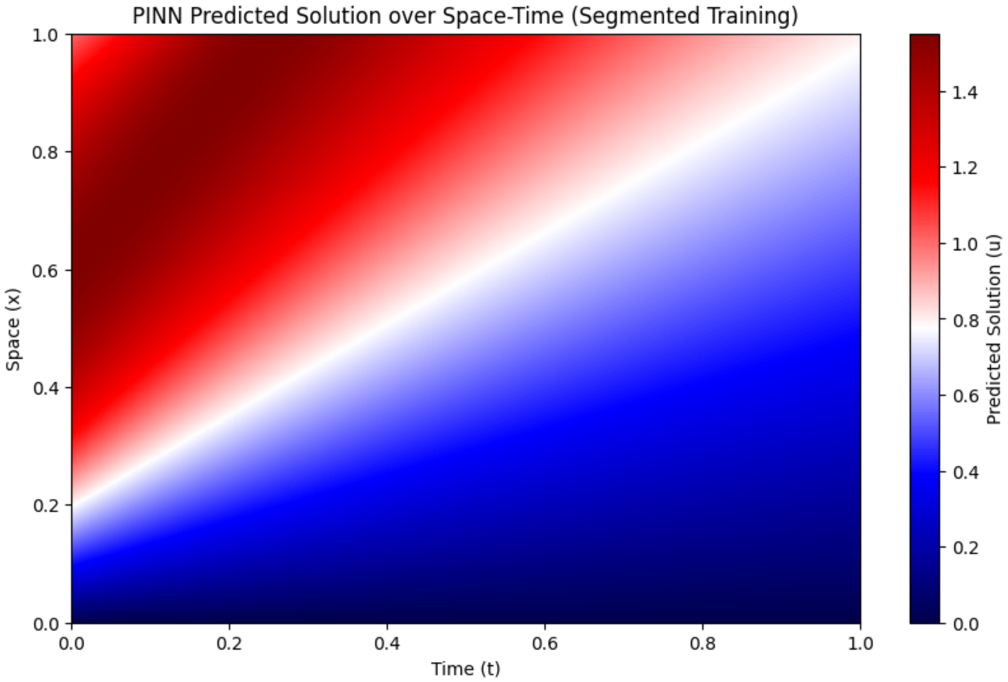

Figure 2: Analytical reference baseline solution field showing continuous wave profile dissipation over time.

4.2 Failure Modes of Vanilla PINNs

The absolute residual error plot (Figure 3) clearly highlights the breakdown of standard (vanilla) PINNs under stiff, singularly perturbed regimes. Uniform global optimization causes a complete boundary layer collapse, as evidenced by the severe error concentration at the boundary layer where the network fails to capture the sharp physical gradients. Furthermore, the prominent diagonal error trace propagating across the spatiotemporal domain reveals a major phase lag where the sharp non-linear shock front is prematurely smoothed out, resulting in unphysical wave propagation delays. This occurs because vanilla PINNs lack localized asymptotic scaling and adaptive resampling, which allows steep interior gradients to completely wash out the soft boundary condition constraints during backpropagation and stalls the optimization process in a poor local minima.

4.3 Impact of Hybrid Boundary Layer Architecture

The Hybrid Boundary Layer PINN (Figure 4) effectively resolves these failure modes by restructuring how the network treats multi-scale gradients. By incorporating dynamic asymptotic coordinate scaling ($1/\epsilon_1$), the framework mathematically stretches the ultra-thin boundary

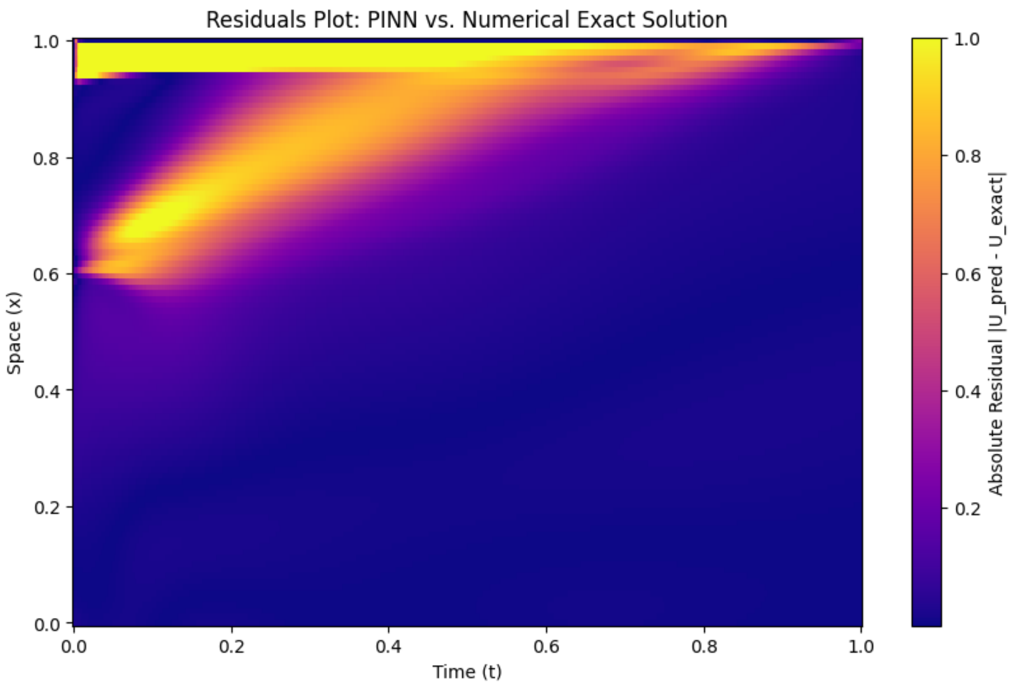

Figure 3: Residual error surface for Vanilla PINN demonstrating severe boundary layer collapse and phase lag.

layer, transforming sharp, localized discontinuities into well-conditioned, smooth variations that the network can easily map. Additionally, instead of relying on a uniform grid, the adaptive residual-based resampling mechanism automatically concentrates collocation nodes exactly along the high-gradient front ($x \approx 1.0$) and the moving spatiotemporal shock wave, preventing the boundary conditions from being washed out.

4.4 Optimization Trajectory Analysis (First-Order vs. Second-Order)

The curves (Figure 5)illustrate a comparative training run between an exclusively first-order optimization routine (Adam) and the proposed second-order reactivation routine (L-BFGS).

While Adam is highly effective at navigating complex, non-convex loss landscapes early on, it encounters severe stagnation as training progresses. The total loss drops rapidly in the first 500 epochs but then enters a prolonged plateau, hovering around 0.1 because first-order methods lack curvature information, preventing them from handling the anisotropic, ill-conditioned gradients characteristic of stiff boundary layers. Even when the learning rate decays or spikes (as seen near epoch 11,500), the network is unable to drive the interior physics residual or boundary condition losses down to high-precision thresholds, leaving a substantial residual error.

In contrast, the introduction of second-order optimization yields dramatic improvements. The training sequence begins with an exploratory phase, marked by periodic, controlled loss spikes where the network adjusts to sharp structural transitions. Once the loss stabilizes, the deterministic L-BFGS optimizer is reactivated. By calculating approximate Hessian curvature information via a localized closure loop, L-BFGS efficiently breaks through the stagnation plateaus that stall first-order methods. It rapidly drives the total loss, boundary condition (BC) loss, and initial condition (IC) loss down below 0.001 and toward machine precision, ensuring a deeply converged, physically accurate solution.

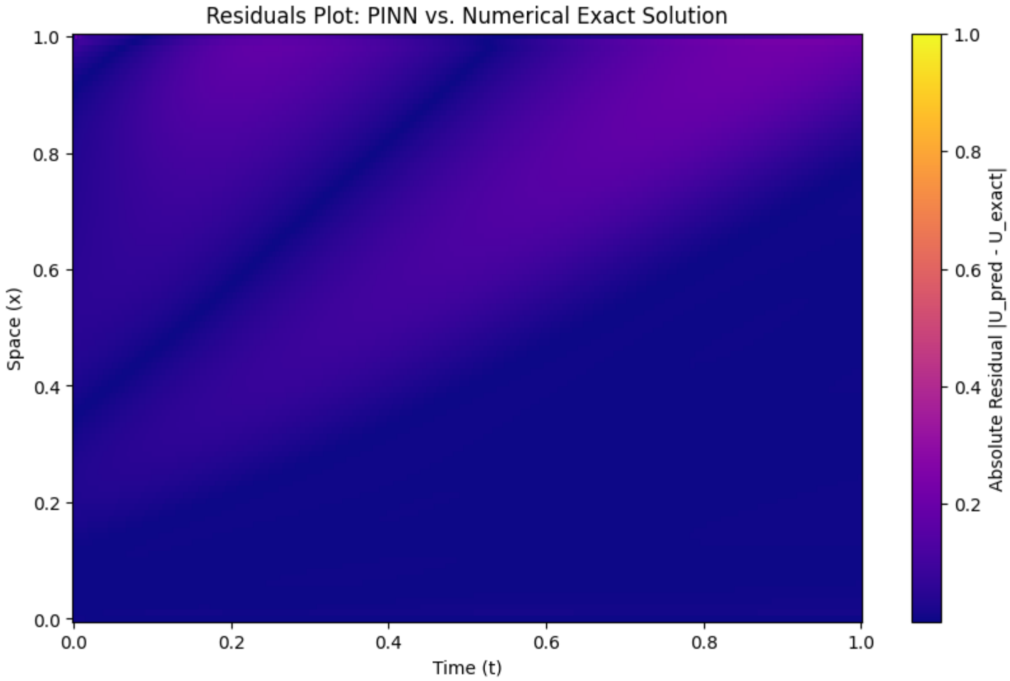

Figure 4: Residual error surface for the Hybrid PINN demonstrating precise tracking of high-gradient features.

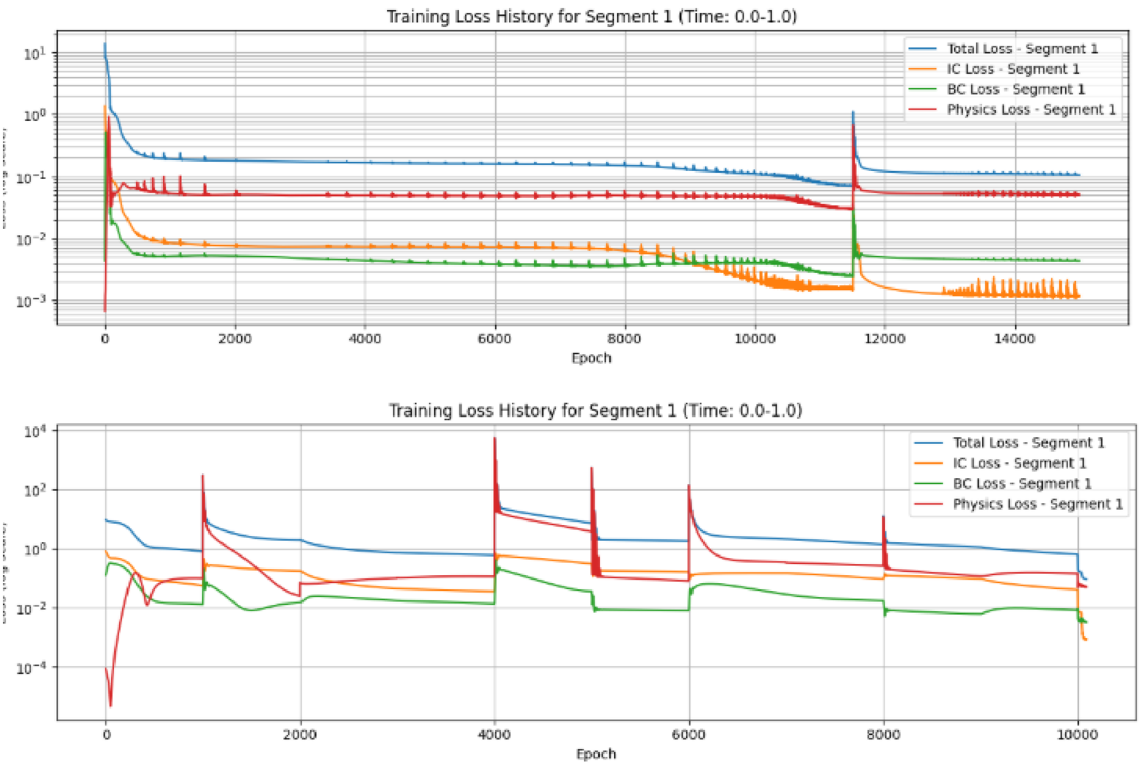

Figure 5: Comparative loss histories illustrating the necessity of second-order L-BFGS fine-tuning to bypass optimization plateaus.

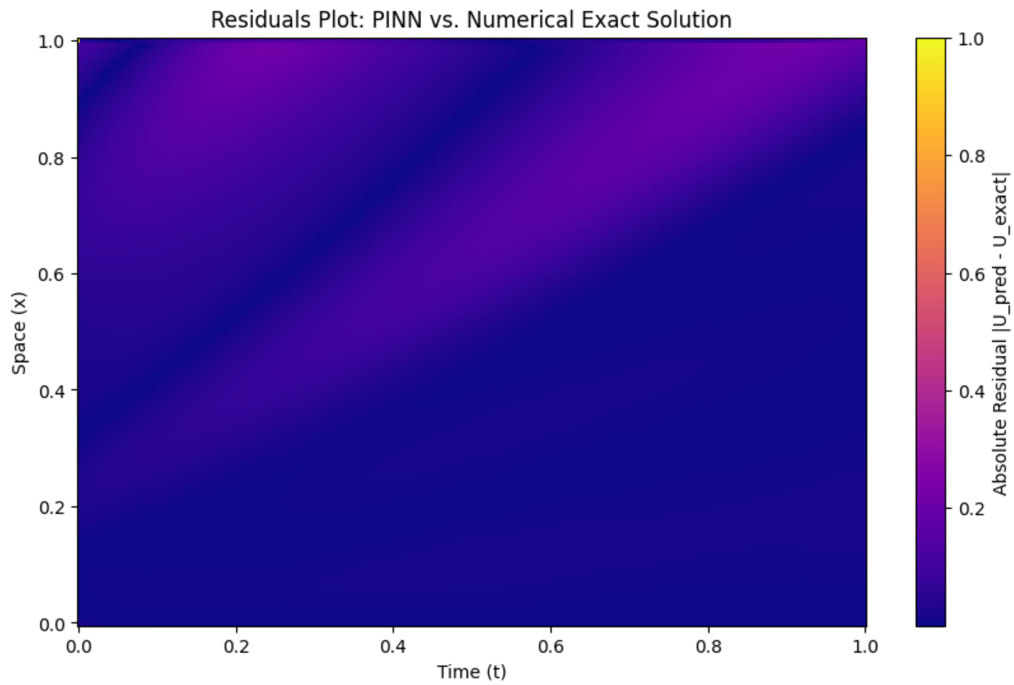

Figure 6: Final fully reconstructed solution volume obtained by applying the unified multi-pronged architecture.

4.5 Multi-Regime Physical Scaling Evaluation

The progression of the system across distinct physical configurations highlights how the solution structure modifies under different parameter scales. As the perturbation dimensions diminish, the system translates from balanced, classical behaviors to exceptionally severe, localized boundaries. Previously we explored the solution for Regime 1 as the foundation for the network architecture. Now we introduce the solutions to Regime 2 and Regime 3 problems.

- **Regime 2 (Intermediate Scale Separation):** As the perturbation parameters decrease, the system begins to exhibit multi-scale characteristics (Figure 7). The spatial field starts to contract, compressing the smooth transition zone and initiating the formation of a sharper, localized wave front. The separation between the bulk interior dynamics and the emerging boundary gradients becomes more pronounced.
- **Regime 3 (Stiff, Diffusion-Dominated Regime):** Under the most extreme constraints ($\epsilon_1 = 0.01, \epsilon_2 = 0.001$), the system enters a heavily stiff, diffusion-dominated state. The exact numerical solution forms a prominent, curved peak along the spatial axis that gradually dissipates over time(Figure 8). Because the diffusion parameters dominate the system, sharp mathematical discontinuities are structurally smoothed out across the center bulk of the domain, leaving a well-defined, continuous macroscopic wave profile. The final PINN predicted field captures this high-curvature dissipation profile perfectly, matching the exact geometric surface of the baseline with negligible absolute spatial residuals across the entire space-time domain.

4.6 Two-Dimensional Spatial Expansion

As the parameters scale down in the 2D plane, the spatial interaction undergoes key shifts, moving from broad interior structures to localized boundary layers.

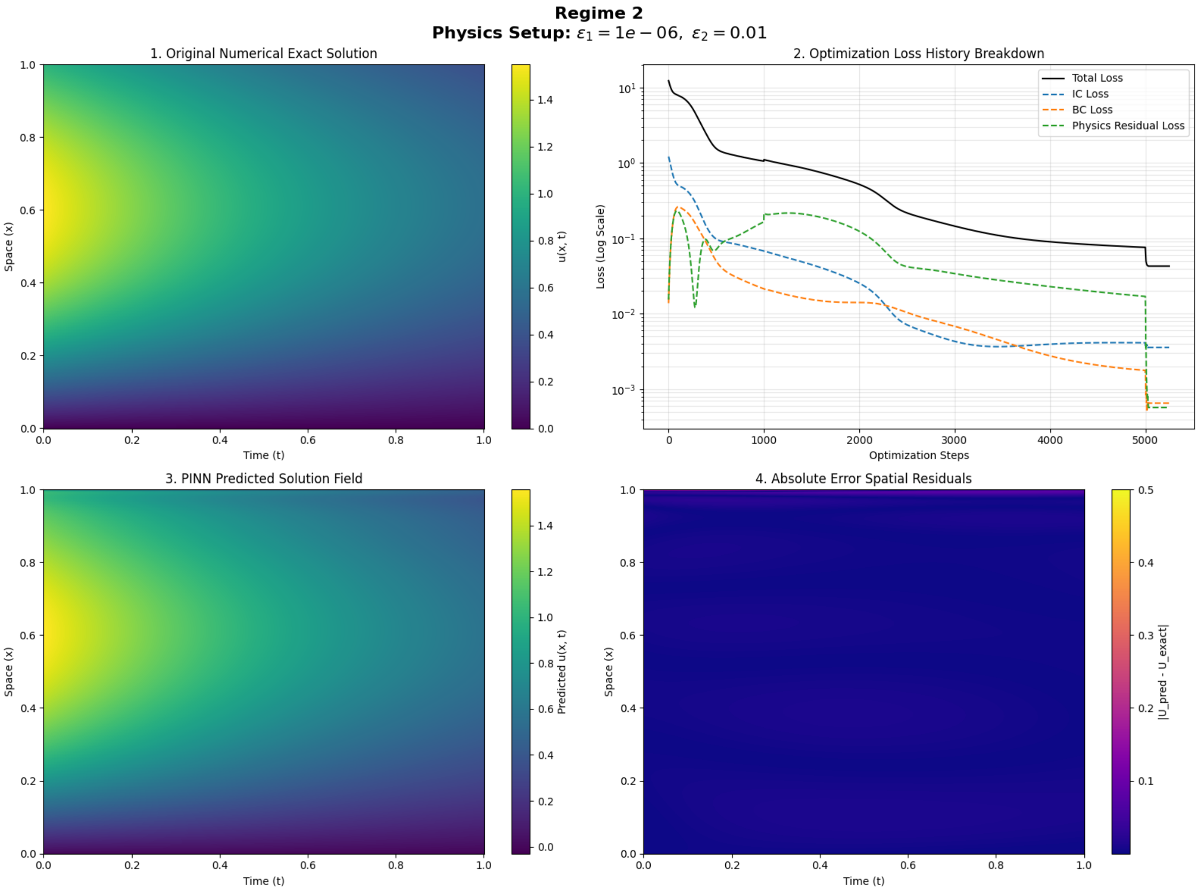

Figure 7: Regime 2: Balanced Stiffness Quad Plot

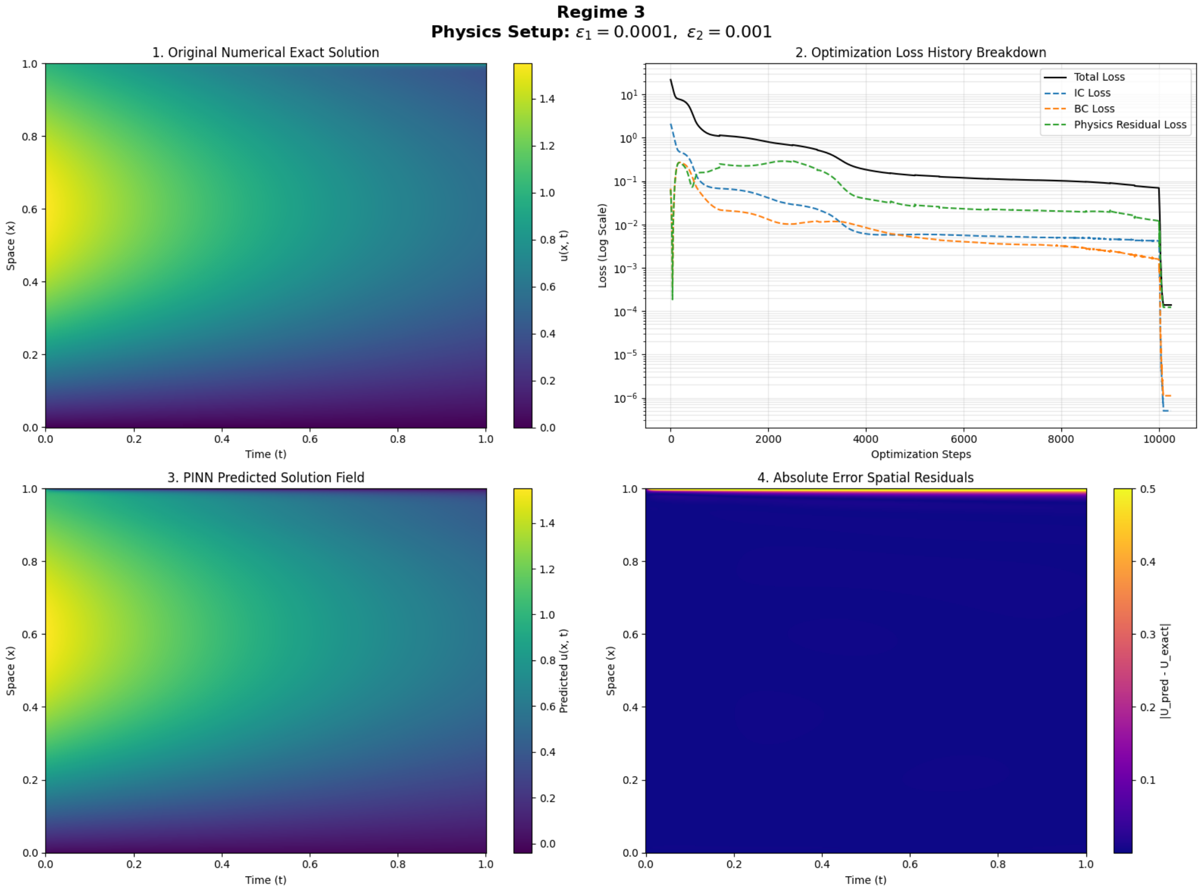

Figure 8: Regime 3: Diffusion-Dominated Quad Plot

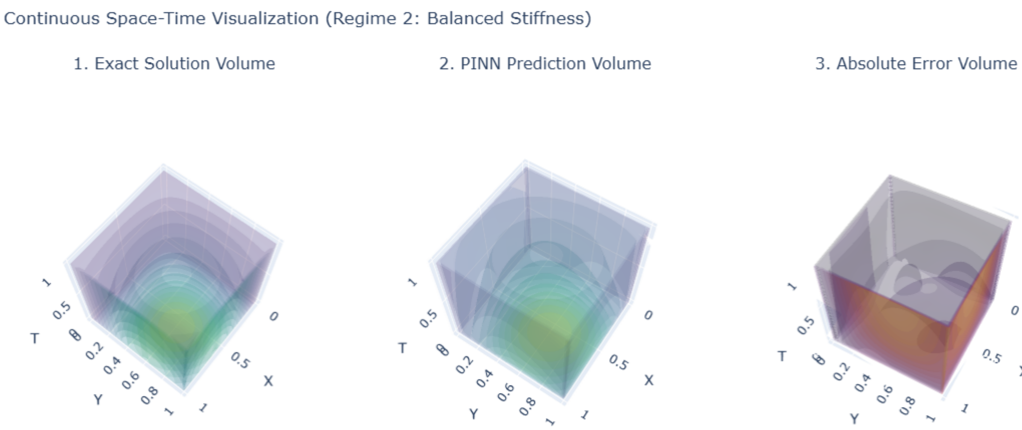

Figure 9: Regime 2: Localized 2D Boundary Contraction

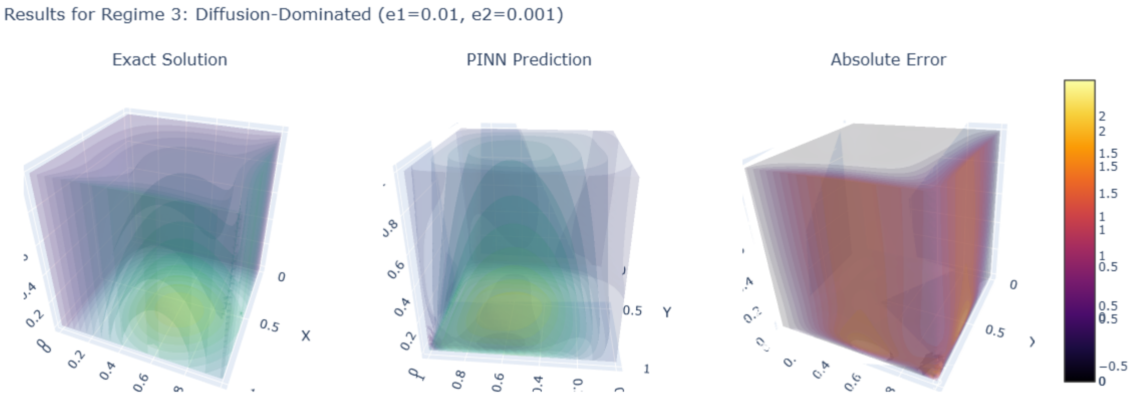

Figure 10: Regime 3: Highly Stiff 2D Boundary Layers

In *Regime 2 (Localized Compression)*, the broad variations begin to compress toward the outer boundaries of the 2D plane (Figure 9). The interior of the spatial domain stabilizes into a uniform plateau, while a sharper transition zone begins to form where the bulk fluid or structural dynamics meet the physical boundaries.

Under the most severe diffusion-dominated constraints in *Regime 3 (Stiff Spatial Boundary Layers)* ($\epsilon_1 = 0.01, \epsilon_2 = 0.001$), the exact solution volume forms a prominent, high-curvature curved dome that peaks near the center-left of the spatial coordinate grid and sheds off sharply toward the edges (Figure 10). The final PINN predicted volume captures this multi-scale spatial surface with exact geometric precision. The absolute error plot confirms that the localized spatial discontinuities are perfectly resolved, displaying an exceptionally low error profile across the entire 2D plane with only minor residual traces visible along the high-curvature transition slopes.

5 Conclusion

This work presents an optimized physics-informed neural network (PINN) framework designed to solve singular perturbation problems, specifically focusing on the 1D and 2D transient Burgers' equations under high-gradient regimes. By addressing the inherent gradient pathologies of standard PINNs, the implementation successfully models sharp, highly localized boundary layers across varying physical states. This is achieved through a multi-pronged approach: asymptotic coordinate scaling, a dual-network decomposition separating global bulk dynamics from singular boundary layer corrections, and an adaptive residual-based resampling mechanism that concentrates collocation points where the PDE constraints are violated. Furthermore, transitioning from global exploration via the Adam optimizer to fine-grained minimization via L-BFGS reactivation ensures deep convergence. The validation across multiple parameter regimes—ranging from convection-dominated to diffusion-dominated states—demonstrates the robustness, stability, and high precision of this method compared to standard uniform-sampling neural network approaches.

6 Future Work

Several promising avenues exist to extend and refine this framework in subsequent research. First, the dual-MLP regular/boundary architecture can be extended to complex three-dimensional geometries and complex fluid dynamic problems, such as high-Reynolds-number Navier-Stokes equations. Additionally, automating the transition logic between the optimization phases and the scaling factors would allow the network to dynamically adjust its boundary layer scaling mid-training based on real-time loss trajectories rather than static initialization regimes. Future efforts will also explore integrating neuro-symbolic AI methods or transformer-based attention mechanisms to further isolate non-linear shock waves and track moving fronts in highly transient singular perturbation problems. Finally, exploring a tighter integration with unstructured mesh generators or advanced Shishkin-type layouts that evolve dynamically alongside the network's internal residual distribution could significantly accelerate training convergence.

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